Investigating the Law of Large Numbers!

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Introduction

When it comes to predicting the weather on a particular day, calculating which basketball superstar is most efficient from the free throw line, or analyzing stocks, chances are you are delving into the world of statistics. It is through this particular branch of mathematics that we are able to calculate risk, predict patterns, and analyze data for a multitude of different reasons.

This paper aims to tackle a specific concept from statistics which is known today as the "Law of Large Numbers" to which we will describe the aforementioned concept, prove the "Weak" Law of Large Numbers and its distinction to the "Strong" Law of Large Numbers. In order to define what the Law of Large Numbers is, we first must take a step back and go over the concept of probability.

Probability

Probability is the likelihood that a particular defined event happens. Perhaps the most recognized example of this is a coin. Given a two sided coin we can extrapolate that for any given flip, the coin will land with either its "head" side or "tail" side and that for any given outcome of the flip, each possibility is assured a 50% chance of occurring. The following definitions are necessary to understand what is to come:

- We define a **population** to be the set of all possible objects/events to be studied. Often times it is difficult to obtain information regarding a population's mean, variance and other such properties (these will be defined below) and so we instead take a subset of obtainable objects/events from this population and measure data from this subset to infer the population's properties. We refer to this as a **sample**. [3] It is often the case that in collecting data for studies such as as average height in a population, or COVID-19 cases in a country, that researchers collect the sufficient sample data to infer the total population's information without needing to survey every person or go through rigorous spending for such a study.
- We say that a set S is the **sample space** containing all outcomes from an event. [1]
  For a single coin flip, we can let S = {H, T} which represents both heads and tails.
- A **random variable** X is a real valued function that illustrates events taken on a sample space. This function can take on a multitude of values that are either countably infinite (discrete) or can take on any real number within a specified range (continuous). We say that $X = x$ where x is the specific value X takes on. In most cases we take our random variable X to be independent or disjoint from any other random variable so as to not influence our dependant variables in the study and so often times we assume independence among our variables. [3]
- We define **probability distribution** as the probability that $X = x$ which is $P(X = x)$ for all x in S in the discrete case and $P(X \leq x)$ for all x such that $-\infty < x < \infty$ in the continuous case. [3] If we let X represent our coin example and let $x = 0$ and $x = 1$ represent the heads and tails case respectively, then: $P(X = 0) = \frac{1}{2}$ and $P(X = 1) = \frac{1}{2}$, where either scenario has a 50% chance of happening.
- The **expected value** or the **mean** E(X) of a random variable X is its average value. This is given by:
$$E(X) = \sum_{y} yp(y)$$
  in the discrete case or:
$$E(X) = \int_{-\infty}^{\infty} yf(y)dy$$
  in the continuous case. We consider the population mean to be μ and the sample mean to be $\bar{X}$. It is important to recognize that μ is not necessarily equal to $\bar{X}$ and for most cases, it is not. [3]

- **Variance**, denoted as $\text{Var}(X)$ for a random variable X, is meant to describe the spread of data relative to the mean. This is given by:

$$\text{Var}(X) = E[(Y - \mu)^2]$$

where every random variable is compared to the mean and it's squared difference is measured. The greater our variance is, the farther apart our data points differ from the given mean. Similar to what was given for expected value, the population variance is described to be σ^2 while the sample mean is s^2. We say that the **standard deviation** is the $\sqrt{\text{Var}(X)} = \sigma$ for the population standard deviation or s for the sample standard deviation. [3]
Lets take this into an example. Presume that we are to study the average height of students in UTM, with an average height of 5'7. A low variance indicates that the students participating in the study are all close to the average (with a majority height lingering around 5'5 - 5'0) whereas a high variance indicates a greater spread (less consistent around the average height of 5'7).

Understanding the Law of Large Numbers

Let $X_1, X_2, \ldots, X_n$ be independent random variables that all follow an identical distribution, each with mean μ and variance σ^2. Suppose we let $\bar{X}$ define the average value of μ across our collection of $X_1, X_2, \ldots, X_n$ such that $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$.

1) **Expected Value**:
To start, is important to note, that because each X_i is of an identical distribution, that each $E(X_i) = \mu$. Then,

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{\sum_{i=1}^n X_i}{n}\right) \\ &= \frac{\sum_{i=1}^n E(X_i)}{n} \\ &= \frac{E(X_1) + E(X_2) + \cdots + E(X_n)}{n} \\ &= \frac{(\mu) + (\mu) + \cdots + (\mu)}{n} \\ &= \frac{n(\mu)}{n} \\ &= \mu \end{aligned} \tag{0.1}$$

which is the expected value of $\bar{X}$.
Next we follow a similar process for the sample variance.

2) **Variance**:
Following a similar result from 1), we have that:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right)$$

So lets simplify! By variance properties, $\text{Var}(aX) = a^2\text{Var}(X)$.

From this, we see that $a = \frac{1}{n}$. Note that as X_i is of identical distribution, that $\mathrm{Var}(X_i) = \sigma^2$. So to follow up on our simplification:

$$\begin{aligned}
&= (\frac{1}{n^2})[\sum_{i=1}^{n} \mathrm{Var}(X_i)] \\
&= (\frac{1}{n^2})[\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n)] \\
&= (\frac{1}{n^2})(\sigma^2 + \sigma^2 + \cdots + \sigma^2) \\
&= \frac{n\sigma^2}{n^2} \\
&= \frac{\sigma^2}{n}
\end{aligned} \tag{0.2}$$

which is the variance for $\bar{X}$.

Now suppose we take the limits of both $E(\bar{X})$ and $\mathrm{Var}(\bar{X})$ from (0.1) and (0.2) respectively:

$$\begin{aligned}
&= \lim_{n \to \infty} E(\bar{X}) \\
&= \lim_{n \to \infty} \mu \\
&= \mu
\end{aligned} \tag{0.3}$$

and,

$$\begin{aligned}
&= \lim_{n \to \infty} \mathrm{Var}(\bar{X}) \\
&= \lim_{n \to \infty} \frac{\sigma^2}{n} \\
&= 0
\end{aligned} \tag{0.4}$$

As $n \to \infty$, what happens for both our expressions?

- As we can see, from (0.3), upon taking larger and larger values of n, our sample mean itself approaches and eventually becomes equal to that of the population mean. This is a very ground-breaking result in the way that it guarantees that our theoretical mean in μ will be produced in our sample, given that we just increase the sample size n.
- From (0.4), we see that as n increases, that $\mathrm{Var}(\bar{X})$ goes to 0. Take note of how we defined variance earlier. Since variance is meant to describe the spread of data from the mean, if $\mathrm{Var}(\bar{X})$ goes to 0, then to put simply, there is no variation in our data points from the mean.

Now lets put all this together into what is finally, the Weak Law of Large Numbers.

3) **Proof**:

To start, lets recall what we have shown so far. We have defined that μ and $\bar{X}$ might not necessarily be the same and that from (0.3) they are equal when $n \to \infty$. So if we gradually increase the value of n then the distance between $\bar{X}$ and μ gets smaller and smaller and approaches 0. Thus, for some $\epsilon > 0$:

$$|\bar{X} - \mu| > \epsilon$$

Next we invoke Russian mathematician Chebyshev and use "Chebyshev's inequality"[2]. His inequality states that:

$$\mathrm{P}(|x - \mu| > k) \leq \frac{\mathrm{Var}(X)}{k^2} \tag{0.5}$$

Chebyshev's inequality is the notion that as we get closer and closer to obtaining the mean (from $|\bar{X} - \mu|$), that a certain number of values exist within this given range, which very interesting to say the least. Since the variance represents the spread, if we decide to observe values closer to the mean, the group of remaining data points gets smaller and smaller. [5] Hence the probability of $|x - \mu| > k$ is always going to be less than or equal to a constant value to which was derived to be $\frac{\mathrm{Var}(X)}{k^2}$. Now by taking the limit and substituting $k = \epsilon$, $x = \bar{X}$, and Var(X)= $\frac{\sigma^2}{n}$,

$$\lim_{n\to\infty} \mathrm{P}(|\bar{X} - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \tag{0.6}$$

Now what do we get here? Similar to (0.4) with taking the limit in regards to sample variance; when we take the limit here, we get that $\frac{\mathrm{Var}(X)}{k^2}$ approaches 0. Thus from (0.6) we conclude:

$$\lim_{n\to\infty} \mathrm{P}(|\bar{X} - \mu| > \epsilon) = 0$$

However, why is it that it is definitely equal to 0? It is crucial to remember that because probability exists from [0,1], that the only $\mathrm{P}(|\bar{X} - \mu| > \epsilon)$ can't be less than 0, therefore $\lim_{n\to\infty} \mathrm{P}(|\bar{X} - \mu| > \epsilon) = 0$. A further consequence of this conclusion implies that:

$$\lim_{n\to\infty} \mathrm{P}(|\bar{X} - \mu| < \epsilon) = 1$$

What we have derived is known more specifically as the "Weak" Law of Large Numbers, which is the implication that the sample average converges to that of the population average given that n gets bigger and bigger. The "Strong" Law of Large Numbers disregards convergence and instead asserts that the sample average <u>is</u> the population average. This would look like:

$$\lim_{n\to\infty} \mathrm{P}(\bar{X}_n = \mu) = 1$$

or

$$\mathrm{P}(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$$

In other words, $\mathrm{P}(\bar{X}_n = \mu) = 1$ shows that it is indeed certain that this is not a convergence but an absolute.

In Conclusion

What does this mean for the world of statistics? The Law of Large Numbers is a beautiful theorem in that it shows how a reoccurring event/trial will always have a set probability/average associated with it. We can use this theorem to make long term statistical inferences on a multitude of events.[1] Look back to our reoccurring coin example, this theorem guarantees that over a large n we are close if not assured that our P(Heads) $\to \frac{1}{2}$. Events such as the probability of drawing a card from the suit of hearts is not only $\frac{1}{4}$ in theory, but in practice will converge to be. This article [4] goes over some very useful applications of the law of large numbers. In betting on different games like Fibber's Dice, one can predict certain outcomes before it even starts by obtaining the expected value of certain hands for card games or certain dice roll combinations. Though such examples here are applied to betting and gambling, this theorem's use can extend far beyond that, into algorithms, into finance estimates and many more!

What do I take from this? Math is a fundamental in nature in that these numbers are meant to be. From conducting a study, is it not incredible that we can perfectly estimate a samples quantitative information before calculating a sufficient number of samples? Math is beautiful and elegant in that we are still discovering new applications of mathematics, new ways of seeing this mathematical world all around us, and it is up to us to continue the pursuit of knowledge!

References

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